🚀 **COURSE NAME: DevOps Bootcamp**

🎯 Section Title: Entering Learning Phase 3 - DevOps Core
🧱 Main Course Section: 8 - Build Automation & CI/CD with Jenkins
🧪 Lab Type: **3 - IAM - Manage Users, Roles and Permissions**

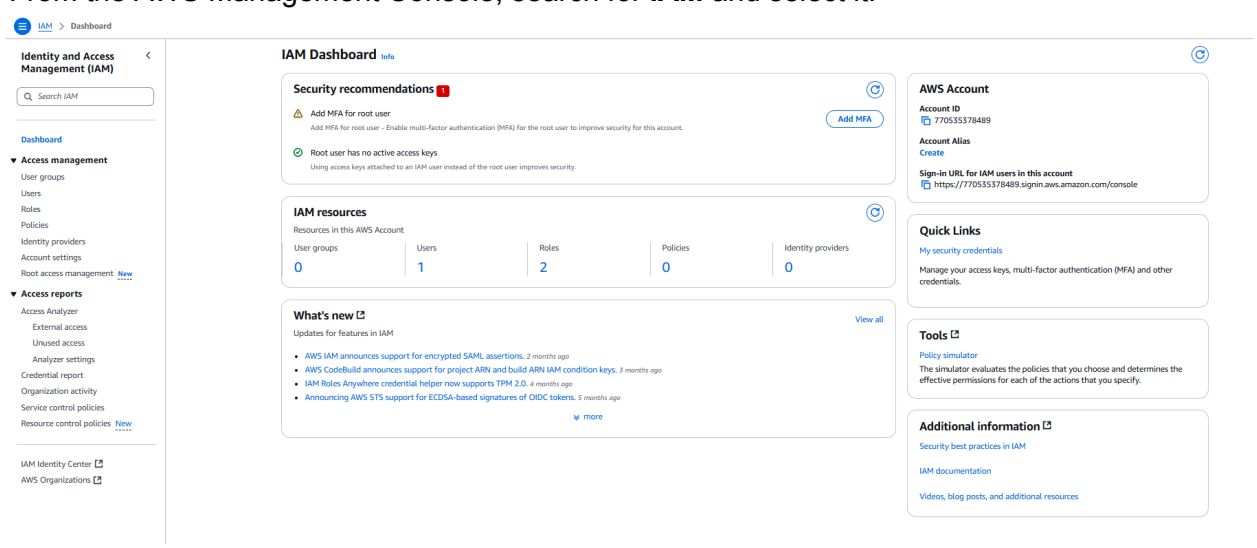
🔐 Step 1: Create an Admin IAM User in AWS with Full Administrative Access

✅ **Objective:**
To securely create an AWS IAM user with full admin privileges so it can be used for Jenkins or other DevOps automation tasks.

🛠️ Steps to Create the Admin User:

1. **Log in to the AWS Console**
   Navigate to: https://console.aws.amazon.com/iam

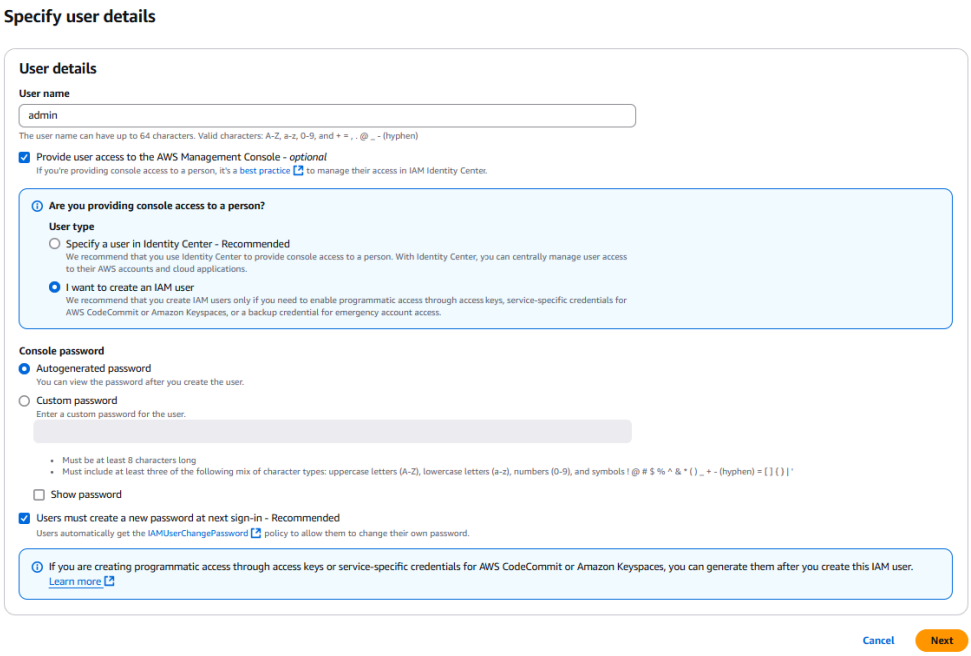
2. **Go to IAM Dashboard**
   From the AWS Management Console, search for **IAM** and select it.



3. **Click on "Users" → "Add users"**

   - Enter a user name (e.g., `jenkins-admin-user`)
   - Select: **Access key – Programmatic access** (for Jenkins or CLI use)

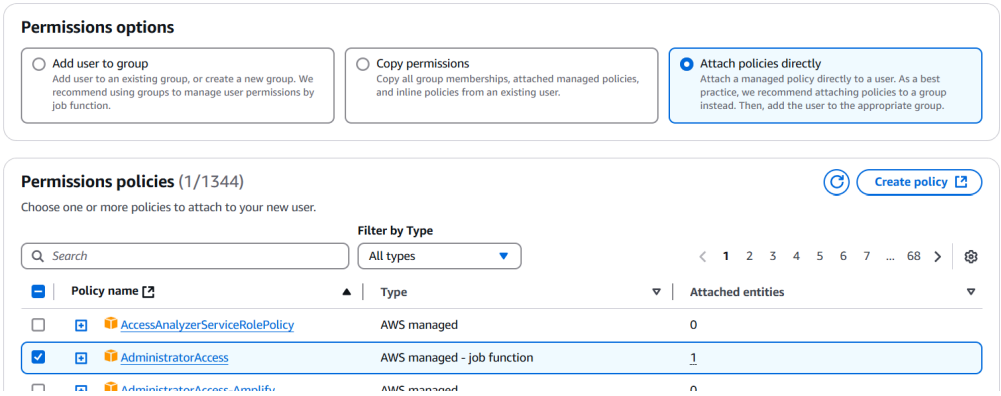
- (Optional) Select **AWS Management Console access** if GUI access is needed



4. **Set Permissions**

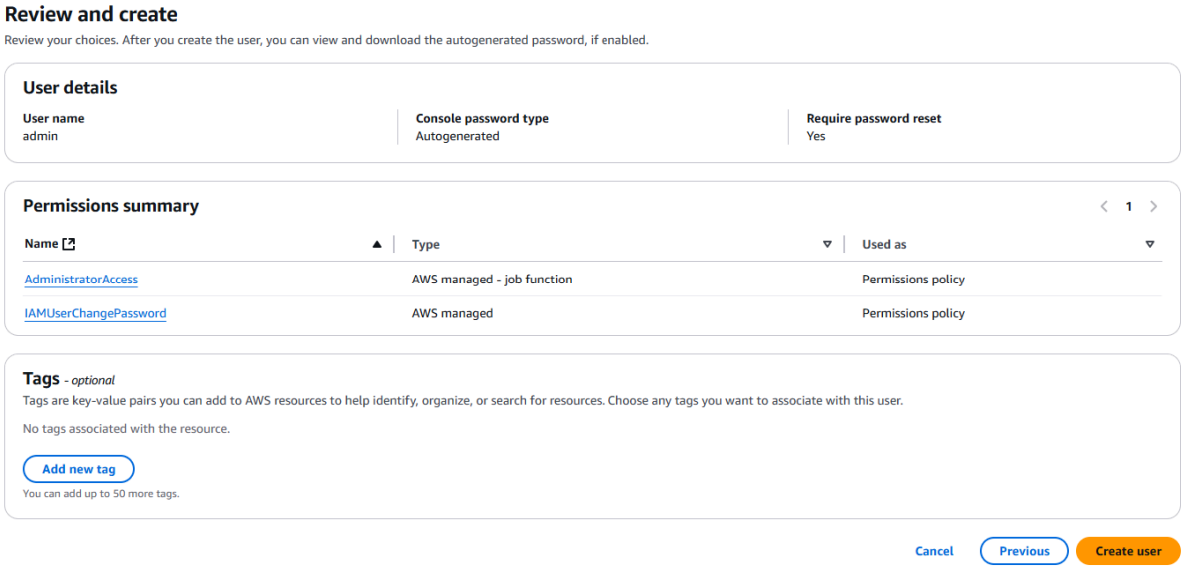
- Choose **Attach policies directly**

- Search and select: **AdministratorAccess**
  💡 *This grants full permissions to all AWS resources.*

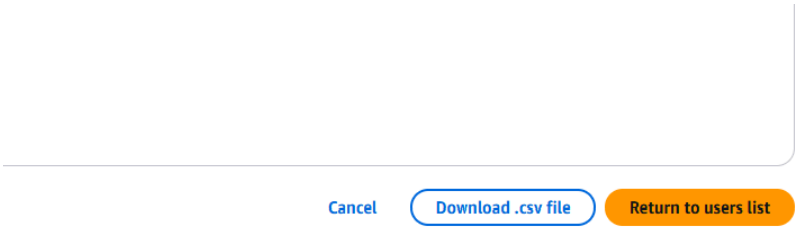


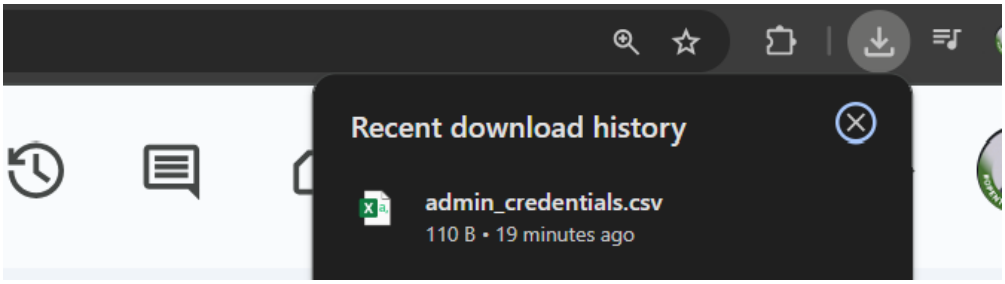
5. **Add Tags (optional)**

- Example: `Key: Project → Value: Jenkins-Shared-Library`



6. **Review and Create**
    - Double-check all inputs
    - Click **Create user**
7. **Save Access Credentials**
    - Download or copy the **Access key ID** and **Secret access key**
      ⚠️ *You'll only see the secret once — save it securely (e.g., password manager or Jenkins credentials).*





✅ **Conclusion:**
In **Step 1**, you successfully created an AWS IAM user with full administrative access. This user is essential for automating infrastructure tasks through Jenkins or the AWS CLI. Saving the access credentials securely ensures your DevOps pipeline can interact with AWS services securely and efficiently. 🔐🚀

🔐 **Step 2: Confirm Console Access for IAM Admin User**

✅ **Objective:**
Verify that your newly created IAM admin user can successfully log in to the AWS Management Console using their username and password.

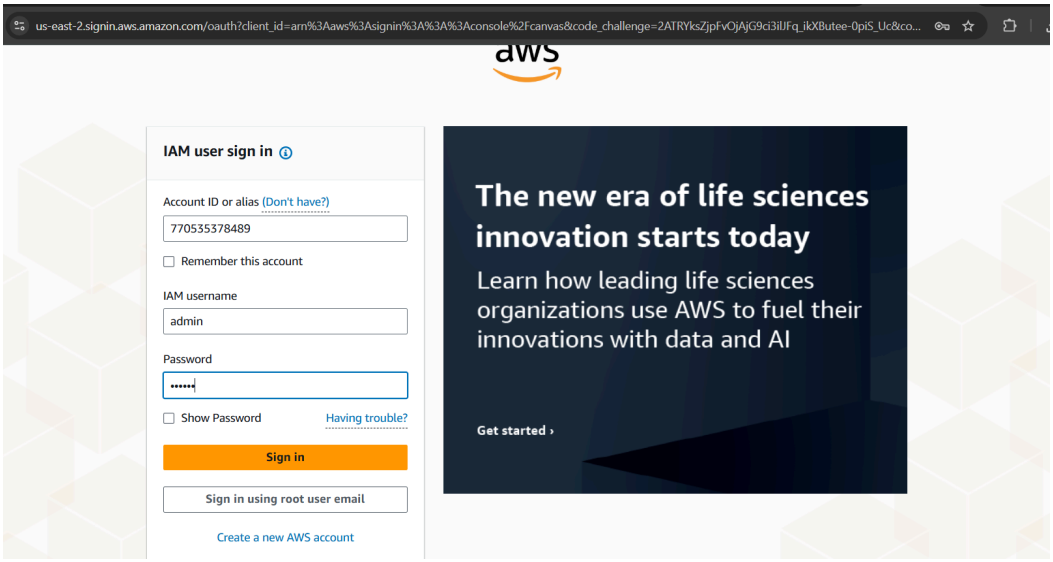
🛠️ **Steps to Confirm Access:**

1. **Go to the IAM Sign-In Page**
   🔗 https://signin.aws.amazon.com/console

2. **Click "Sign in as IAM user"**

   - Enter your **AWS Account ID or alias**

- Enter the **IAM username** (e.g., `admin-user`)



- Enter the **password** set during user creation

3. **Verify Access to AWS Console**

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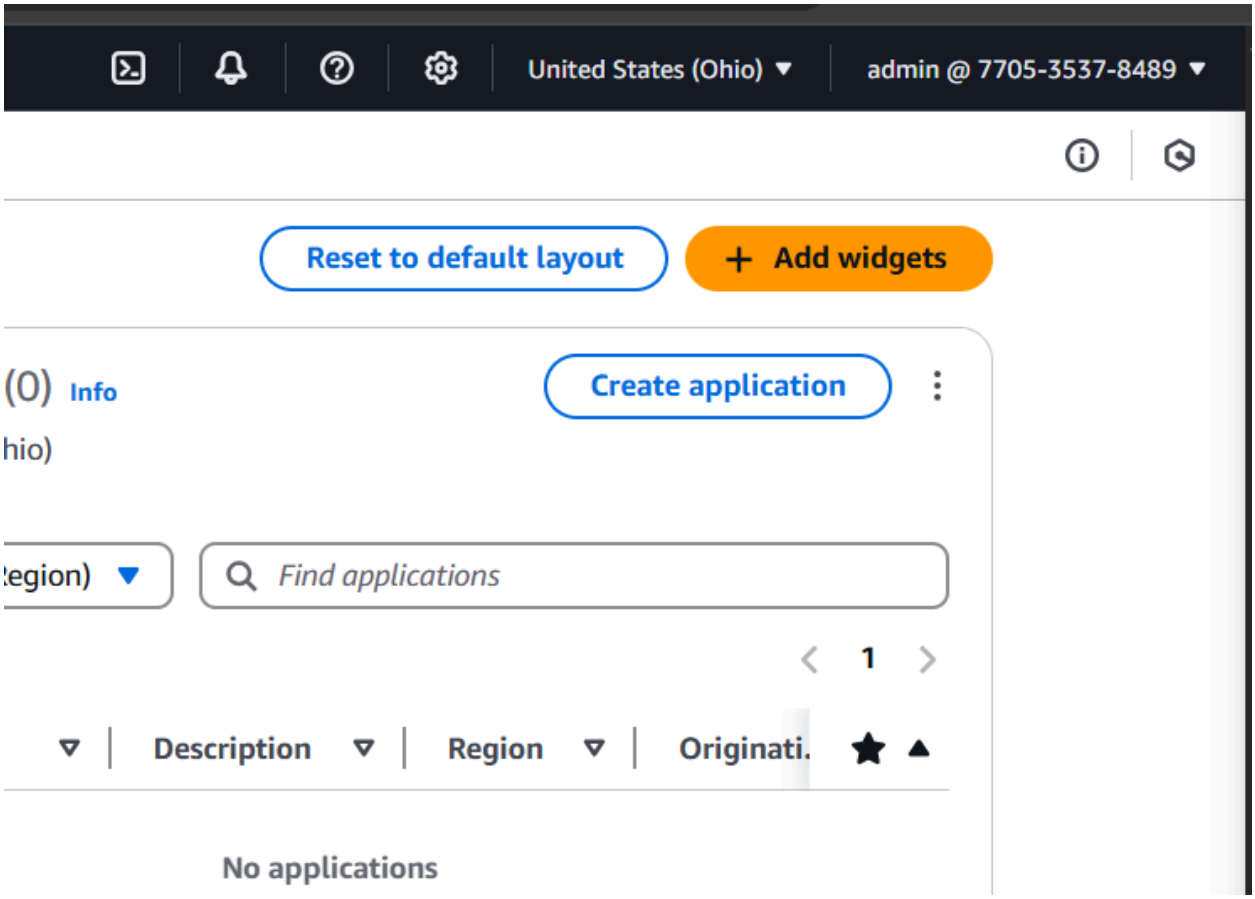


nce signed in:

- Navigate to services like **EC2**, **IAM**, and **S3**
- Confirm the user can view and manage these services, proving the admin permissions are working

✅ **Success!**
Once you're logged in and confirmed access works, you're ready to move on to your next AWS setup step — whatever that may be! ☁️✅

⚙️ Step 3: Generate IAM Admin Access Keys for Future Use (e.g., Jenkins Integration)
✅ Objective:

To generate and download **Access Key ID** and **Secret Access Key** for the IAM admin user created in Step 1 — to be used later in Jenkins or AWS CLI.

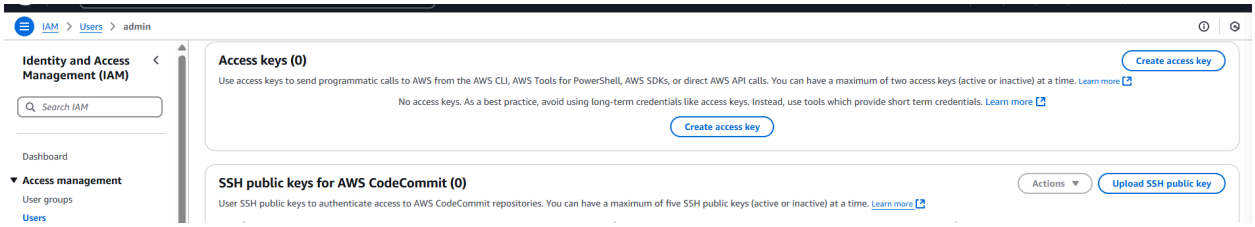
🛠️ Steps to Create and Download Access Credentials:

1. **Log into AWS Console**
   → https://console.aws.amazon.com/iam

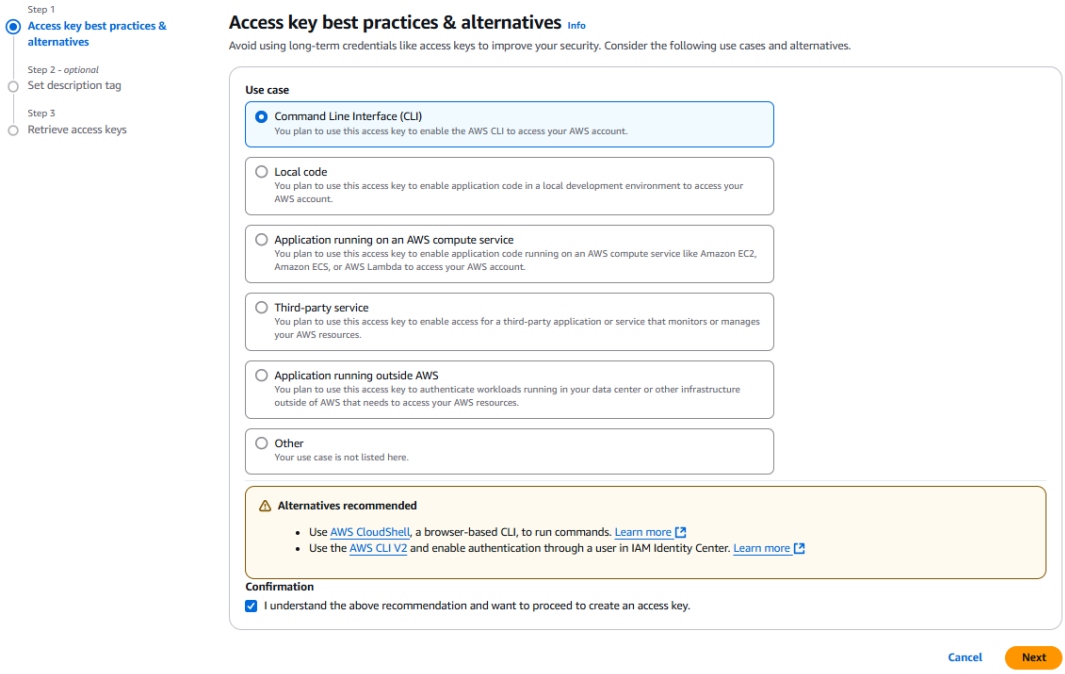
2. **Navigate to IAM → Users → Select Your Admin User**
   (e.g., `jenkins-admin-user`)

3. **Click the "Security credentials" tab**

4. **Scroll to "Access keys" → Click "Create access key"**

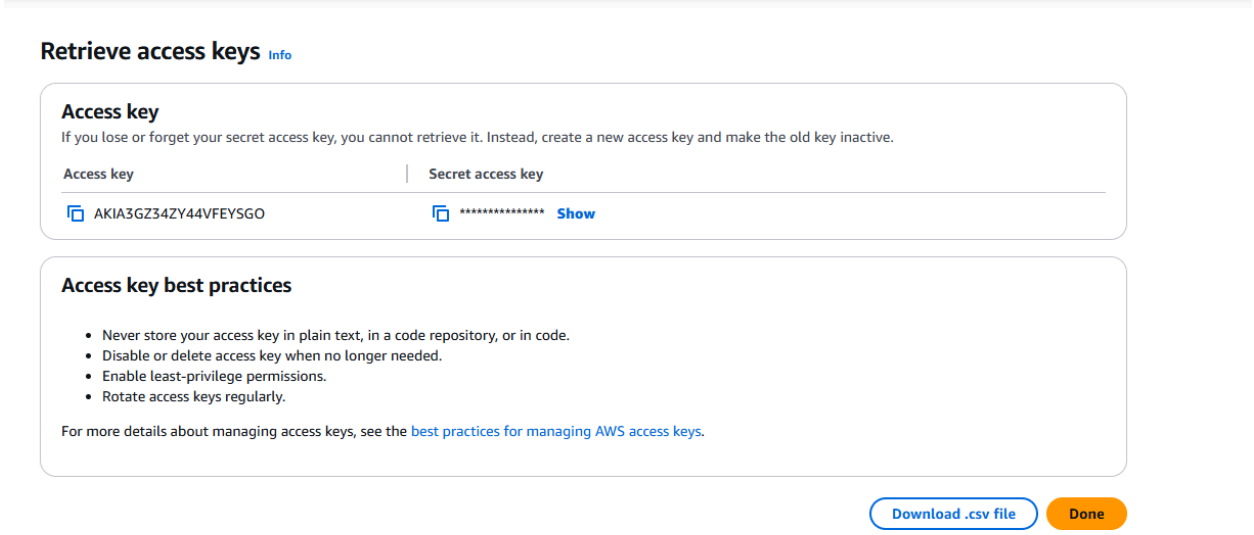


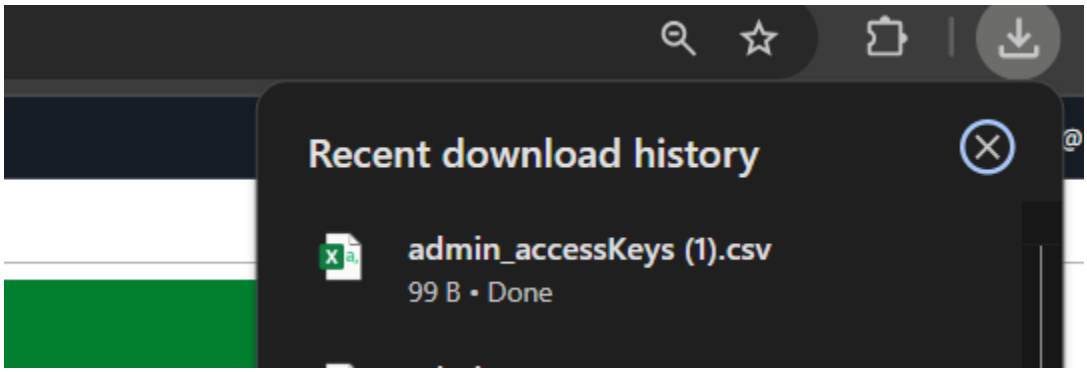
5. **Select “Command Line Interface (CLI)” as use case**
   → Click **Next**, then **Create access key**



6. **Download the `.csv` file** or **copy** both:

   - **Access Key ID**
   - **Secret Access Key**





7. ⚠️ *Note: This is your only chance to view the secret key. Store it securely (e.g., password manager or Jenkins credentials manager).*

✅ **Conclusion:**
In **Step 3**, you've created secure programmatic credentials for your IAM admin user. These will be essential for tools like Jenkins or CLI access, but you're not configuring anything yet — just generating and storing them for safe use later. 🔐📄✅